INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Elk Grove Village, Illinois -- station KORD, America/Chicago
Committed pair	OSM 863162820 -> 377032061 Stream Chicago II / Equinix Chicago CH3
Facade gap	118.4 m between the two halls
Weather record	43,775 real hourly records from KORD
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.4108 C at 240 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2025-08-30 (recirc_edge)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 24 hours with 2 mode changes. That is 1 more than the reactive incumbent, at 0 unsafe hours against its 0. 6 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	8 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
	6 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	18.215	21.0	17.780	1.071
01	FREE	yes	-	18.197	21.0	17.780	1.005
02	FREE	yes	-	18.199	21.0	17.780	0.970
03	FREE	yes	-	18.228	21.0	17.780	0.935
04	FREE	yes	-	18.245	21.0	17.780	0.895
05	FREE	yes	-	18.265	21.0	17.780	0.820
06	FREE	yes	-	18.725	21.0	18.330	0.770
07	FREE	yes	-	20.019	21.0	18.330	1.468
08	mechanical	no	dry-bulb	21.176	21.0	18.890	1.731
09	mechanical	no	dry-bulb	22.352	21.0	19.440	1.842
10	mechanical	no	dry-bulb	22.965	21.0	20.560	1.779
11	mechanical	no	dry-bulb	22.874	21.0	21.110	1.325
12	mechanical	no	dry-bulb	23.158	21.0	21.670	1.366
13	mechanical	no	dry-bulb	24.010	21.0	22.613	1.336
14	mechanical	no	dry-bulb	23.192	21.0	22.220	0.982
15	mechanical	no	dry-bulb	22.967	21.0	21.670	1.108
16	mechanical	no	dry-bulb	22.639	21.0	21.110	1.152
17	mechanical	no	dry-bulb	21.791	21.0	20.560	0.952
18	mechanical	yes	switch budget	20.952	21.0	20.000	0.982

19	mechanical	yes	switch budget	20.098	21.0	18.890	1.173
20	mechanical	yes	switch budget	19.634	21.0	18.330	1.285
21	mechanical	yes	switch budget	18.968	21.0	17.780	1.042
22	mechanical	yes	switch budget	18.416	21.0	17.780	0.882
23	mechanical	yes	switch budget	18.178	21.0	17.220	1.081

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.215 C, which is 2.785 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.071 C, and the margin is measured, not chosen: 0.912 C of group-conditional forecast error for this hour of day, plus 0.1599 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.197 C, which is 2.803 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.005 C, and the margin is measured, not chosen: 0.863 C of group-conditional forecast error for this hour of day, plus 0.1418 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.199 C, which is 2.801 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.970 C, and the margin is measured, not chosen: 0.825 C of group-conditional forecast error for this hour of day, plus 0.1442 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.228 C, which is 2.772 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.935 C, and the margin is measured, not chosen: 0.762 C of group-conditional forecast error for this hour of day, plus 0.1729 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.245 C, which is 2.755 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.895 C, and the margin is measured, not chosen: 0.710 C of group-conditional forecast error for this hour of day, plus 0.1853 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.265 C, which is 2.735 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.820 C, and the margin is measured, not chosen: 0.615 C of group-conditional forecast error for this hour of day, plus 0.2052 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.725 C, which is 2.275 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.770 C, and the margin is measured, not chosen: 0.661 C of group-conditional forecast error for this hour of day, plus 0.1097 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.019 C, which is 0.981 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.467 C, and the margin is measured, not chosen: 1.225 C of group-conditional forecast error for this hour of day, plus 0.2424 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.176 C against a 21.0 C limit, so it fails by 0.176 C. It would take a limit 0.176 C higher, or a bound 0.176 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.352 C against a 21.0 C limit, so it fails by 1.352 C. It would take a limit 1.352 C higher, or a bound 1.352 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.965 C against a 21.0 C limit, so it fails by 1.965 C. It would take a limit 1.965 C higher, or a bound 1.965 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.874 C against a 21.0 C limit, so it fails by 1.874 C. It would take a limit 1.874 C higher, or a bound 1.874 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.158 C against a 21.0 C limit, so it fails by 2.158 C. It would take a limit 2.158 C higher, or a bound 2.158 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.010 C against a 21.0 C limit, so it fails by 3.010 C. It would take a limit 3.010 C higher, or a bound 3.010 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.192 C against a 21.0 C limit, so it fails by 2.192 C. It would take a limit 2.192 C higher, or a bound 2.192 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.967 C against a 21.0 C limit, so it fails by 1.967 C. It would take a limit 1.967 C higher, or a bound 1.967 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.639 C against a 21.0 C limit, so it fails by 1.639 C. It would take a limit 1.639 C higher, or a bound 1.639 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.791 C against a 21.0 C limit, so it fails by 0.791 C. It would take a limit 0.791 C higher, or a bound 0.791 C tighter, to change this hour.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.952 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.098 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.634 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.968 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.417 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.178 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

WHAT IT IS WORTH OVER FIVE REAL YEARS, AT THIS SITE

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Held-out days simulated   913 -- the agent never calibrates on a day it is scored on
Free cooling delivered    16.48 h/day, i.e. 6,018 h/yr
Chiller-hours avoided     +326.8 h/yr against a tuned reactive incumbent
12-hour plan stability    94.6 % of 21,882 re-plans change nothing at all
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The ladder, one constraint at a time:

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   N-56-like: notice 0, skill 1.00, no constraint    +57.2 h/yr
   + switch budget 2, min dwell 3 h                 +103.2 h/yr
   + dew-point gate 15 C (Green Grid WP#46 p.6)     +104.4 h/yr
   + notice 3 h, skill 0.50 (no perfect forecast)   +326.8 h/yr
   + unanchored, 4 measured FG offsets rotated      -147.6 h/yr
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PRICED, IN THIS STATE'S OWN ELECTRICITY TARIFF

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Chiller power         156.1 kW per MW of IT load (0.549 kW/ton, the ASHRAE 90.1-2019 minimum)
Electricity           11.81 cents/kWh (Illinois commercial, 2024 annual)
Energy avoided        51,022 kWh per MW of IT load per year
Value                 $6,026 per MW of IT load per year
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Everything above is PER MEGAWATT OF IT LOAD. This project has never measured a data centre's size and will not invent one; a reader who knows their own IT load multiplies once.

WHAT IS NOT CLAIMED

- The chiller COMPRESSOR only. Fans, chilled-water pumps, condenser pumps and cooling-tower fans keep running, and an airside economizer moves MORE air, so fan power can rise. The unmeasured term has the OPPOSITE SIGN, so this is an upper bound.
- Code-minimum chiller efficiency is the OPTIMISTIC end: Standard 90.1 is a floor, hyperscale plants beat it, and a better chiller saves less per hour switched off.
- Full-load kW/ton OVERSTATES the draw at the moment free cooling is available, because free cooling happens when it is cool and a chiller at a cool condenser runs below its rating. IPLV is the lower published number but is an annual-weighted metric under AHRI's assumed load profile, not the part-load point free-cooling weather sits at.
- The heat rejected is APPROXIMATED by the IT load. UPS, PDU and lighting losses inside the envelope add to it; some overhead sits outside the cooled space, so scaling by full PUE would overshoot. No PUE multiplier is applied.
- EIA STATE-SECTOR AVERAGES, not the site's tariff. A Loudoun County campus buys on a large-general-service contract that is not public.
- No carbon, no water, no maintenance and no demand charges. Tower water use moves the OTHER way when the chiller runs less.
- Every caveat on the hours is inherited: the headline is conditional on the bank sitting on the long facade (the refusal guard is priced at -3,124 h/yr where it fires) and the unanchored row is NEGATIVE. Those rows appear here with their signs intact.

HOW TO REPRODUCE EVERY NUMBER IN THIS REPORT

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cd INTAKE-ARBITER/src && python run_all.py
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That rebuilds every artefact from saved data and then audits it: 39 checks, 68 published figures re-read out of the files the code itself wrote, and it exits non-zero on any failure. It makes ZERO API calls -- every input is a saved FortyGuard response, a committed geometry file, or the station's own hourly record.

Generated by INTAKE-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.